EXPERIMENT-2

AIM:- To convert the given ER diagram of an e-commerce system into a relational schema and implement it in MYSQL using CREATE TABLE statements with Primary Keys, Foreign Keys, NOT NULL, UNIQUE, ON DELETE CASCADE and ON DELETE SET NULL constraints. To insert sample records and demonstrate referential integrity voilations.

Relational Schema

The ER diagram can be converted into the following relational schema:

```
CUSTOMER (
    CustomerID PK,
    Name,
    Email UNIQUE,
    Password,
    DOB;
)
```

CUSTOMER_PHONE (
    CustomerID PK, FK,
    Phone PK
)

CUSTOMER_ALT_PHONE (
    CustomerID PK, FK,
    AltPhone PK
)

SELLER (
    SellerID PK,
    SellerName,
    Email UNIQUE,
    Phone,
    GSTIN UNIQUE,
    AccountNo,
    BankName,
    IFSCCode
)

CATEGORY (
    CategoryID PK,
    CategoryName,
    Description,
    ParentCategoryID FK
)

PRODUCT (
    ProductID PK,
    ProductName,
    Description,

Price,
StockQty,
Weight,
SellerID FK,
CategoryID FK
)

PRODUCT_IMAGE (
ProductID PK, FK,
Image PK
)

ELECTRONICS (
ProductID PK, FK,
Brand,
Warranty,
PowerRating
)

CLOTHING (
ProductID PK, FK,
Size,
Color,
Material
)

BOOKS (
ProductID PK, FK,
ISBN UNIQUE,
Author,
Publisher
)

ORDERS (
OrderID PK,
CustomerID FK,
OrderDate,

```
    TotalAmount,
    OrderStatus
)

ORDER_COUPON (
    OrderID PK, FK,
    CouponCode PK
)

ORDER_ITEM (
    OrderID PK, FK,
    ItemNo PK,
    ProductID FK,
    Quantity,
    ItemPrice,
    Discount,
    TaxAmount
)

PAYMENT (
    PaymentID PK,
    OrderID FK,
    TransactionID UNIQUE,
    PaymentDate,
    Amount,
    PaymentMethod
)

DELIVERY (
    DeliveryID PK,
    OrderID FK,
    AddressID FK,
    TrackingID UNIQUE,
    DeliveryStatus,
    DeliveryDate,
    ShippingCharge
)
```

```
ADDRESS (
    AddressID PK,
    CustomerID FK,
    DoorNo,
    Area,
    City,
    State,
    PinCode,
    AddressType
)

ADDRESS_MOBILE (
    AddressID PK, FK,
    MobileNo  PK
)
```

Important Conversion points

- OrderItem is a weak entity, therefore its primary key is (OrderID, ItemNo).
- Customer.Phone, Customer.AltPhone, Product.Image, Order.CouponCodes, and Address.MobileNo are multivalued attributes, so separate relations are created.
- BankDetails is a composite attribute of Seller, so its components are stored directly in SELLER.
- Electronics, Clothing, and Books subclasses of Product, so each uses ProductID as both PK and FK.

- ParentCategoryID is self-referencing and uses ON DELETE SET NULL.
- Deleting a customer cascades to related addresses, orders and dependent records.

Source Code

Run the following code in MySQL 8.x.

```
CREATE DATABASE IS NOT EXISTS DBMS_Lab_file;
USE DBMS_Lab_File;
-- 1. CUSTOMER

   CREATE TABLE Customer(
        CustomerID INT PRIMARY KEY,
        Email VARCHAR(100) NOT NULL UNIQUE,
        Name VARCHAR(50) NOT NULL,
        Password VARCHAR(100) NOT NULL,
        DOB DATE NOT NULL
   );
-- Multivalued attribute: Phone
CREATE TABLE Customer_Phone(
    CustomerIDINT NOT NULL,
    Phone VARCHAR(15) NOT NULL,
    PRIMARY KEY (CustomerID, Phone),

    FOREIGN KEY (CustomerID)
        REFERENCES Customer(CustomerID)
        ON DELETE CASCADE
);
-- Multivalued attribute: AlternatePhone.
   CREATE TABLE Customer_AltPhone(
        CustomerID INT NOT NULL,
        AltPhone VARCHAR(15) NOT NULL,
        PRIMARY KEY (CustomerID, AltPhone),
```

```
    FOREIGN KEY (CustomerID)
        REFERENCES Customer(CustomerID)
        ON DELETE CASCADE
);

--2. SELLER.

CREATE TABLE Seller(
    SellerID INT PRIMARY KEY,
    SellerName VARCHAR(60) NOT Null,
    Email VARCHAR(100) NOT NULL UNIQUE,
    Phone VARCHAR(15) NOT NULL,
    GSTIN VARCHAR(20) NOT NULL UNIQUE,
    AccountNo VARCHAR(30) NOT NULL,
    BankName VARCHAR(60) NOT NULL,
    IFSCode VARCHAR(15) NOT NULL.
);

-- 3. CATEGORY
CREATE TABLE Category(
    CategoryID INT PRIMARY KEY,
    CategoryName VARCHAR(60) NOT NULL,
    Description VARCHAR(200),
    ParentCategoryID INT NULL,

    FOREIGN KEY (ParentCategoryID)
        REFERENCES Category(CategoryID)
        ON DELETE SET NULL
);

--4. PRODUCT

CREATE TABLE Product(
    ProductID INT PRIMARY KEY,
    ProductName VARCHAR(100) NOT NULL,
    Description VARCHAR(255),
    Price DECIMAL(10,2) NOT NULL,
    StockQty INT NOT NULL,
    Weight DECIMAL(8,2),
    SellerID INT NULL,
    CategoryID INT NOT NULL,
```

```
    FOREIGN KEY (SellerID)
        REFERENCES seller (SellerID)
        ON DELETE SET NULL,

    FOREIGN KEY (categoryID)
        REFERENCES category (categoryID)
        ON DELETE CASCADE,
    CHECK (Price >=0),
    CHECK (StockQty >=0)
);

-- 5. ADDRESS TABLE

  CREATE TABLE Address (
      address_id INT PRIMARY KEY,
      customer_id INT NOT NULL,
      street VARCHAR(100) NOT NULL,
      city VARCHAR(50) NOT NULL,
      state VARCHAR(50) NOT NULL,
      pincode VARCHAR(10) NOT NULL,

    CONSTRAINT fk_address_customer
        FOREIGN KEY (customer_id)
        REFERENCES Customer (customer_id)
        ON DELETE CASCADE
);

-- 6. ORDERS TABLE

  CREATE TABLE Orders (
      order_id INT PRIMARY KEY,
      customer_id INT NOT NULL,
      order_date DATE NOT NULL,
      total_amount DECIMAL(10,2) NOT NULL,

    CONSTRAINT fk_order_customer
        FOREIGN KEY (customer_id)
        REFERENCES Customer (customer_id)
        ON DELETE CASCADE
);
```

--7. PAYMENT TABLE

```
CREATE TABLE Payment (
    payment_id INT PRIMARY KEY,
    order_id INT NOT NULL UNIQUE,
    payment_date DATE NOT NULL,
    amount DECIMAL(10,2) NOT NULL,
    payment_mode VARCHAR(30) NOT NULL,

    CONSTRAINT fk_payment_order
        FOREIGN KEY (order_id)
        REFERENCES Orders(order_id)
        ON DELETE CASCADE
);
```

--8. DELIVERY TABLE

```
CREATE TABLE Delivery (
    delivery_id INT PRIMARY KEY,
    order_id INT NOT NULL UNIQUE,
    delivery_date Date,
    status VARCHAR(30) NOT NULL,

    CONSTRAINT fk_delivery_order
        FOREIGN KEY (order_id)
        REFERENCES Orders (order_id)
        ON DELETE CASCADE
);
```

--9. ~~SELLER TABLE~~

```
CRETE TABLE Order_Item (
    order_id INT,
    item_no INT,
    product_id INT NOT NULL,
    quantity INT NOT NULL,
    unit_price DECIMAL(10,2) NOT NULL,

    PRIMARY KEY (order_id, item_no),
```

```
    CONSTRAINT fk_item_order
        FOREIGN KEY (order_id)
        REFERENCES orders (order_id)
        ON DELETE CASCADE,

    CONSTRAINT fk_item_product
        FOREIGN KEY (product_id)
        REFERENCES Product (product_id)
        ON DELETE CASCADE
);
```

Sample Data

Run the following INSERT statements in this order because foreign-key tables depend on parent tables.

```
-- Customer DATA

INSERT INTO Customer
VALUES
(101, 'Aradhana Verma', 'ardhana@gmail.com', 'ara123', '2008-08-13'),
(102, 'Riya', 'riya@gmail.com', 'riya123', '2006-11-16'),
(103, 'Shatakshi', 'shatakshi@gmail.com', 'sata123', '2006-02-16');

INSERT INTO Customer_Phone
VALUES
(101, '9123456780'),
(102, '9988776655'),
(103, '9090909086');

INSERT INTO Customer_AltPhone
VALUES
    (101, '9000011111'),
    (102, '9000022222');
```

```
--SELLER DATA

INSERT INTO seller
VALUES
(201, 'Tech World', techworld@gmail.com', '9876500001',
'GSTIN201A', '1234567890', 'SBI', 'SBIN0001234'),
(202, 'FashionHUB', fashionhub@gmail.com', '9876500002',
'GSTIN202B', '2345678901', 'HDFC', 'HDFC0002345');

-- CATEGORY DATA
INSERT INO Category
(CategoryID, CategoryName, Description, ParentCategoryID)
VALUES
(1, 'Electronics', 'Electronic product', NULL),
(2, 'Mobile Phones', 'Smartphones and mobile devices', 1),
(3, 'Clothing', 'Clothes and fasion products', NULL),
(4, 'Books', 'Books and study material', NULL);

-- PRODUCT DATA
INSERT INTO Product
VALUES
(301, 'Samsung Galaxy A55', '5G Smartphone',
34999.00, 20, 0.21, 201, 2),
(302, 'Denim Jacket', 'Blue denim jacket', '2499.00, 15, 0.08, 2023),
(303, 'DBMS Fundamentals', 'Database management book',
699.00, 30, 0.50, NULL, 4);

-- PRODUCT IMAGE
INSERT INO Pooduct Image
VALUES
(301, 'samsung-a55.jpg'),
(301, 'samsung-a55-back.jpg'),
(302, 'denim_jacket.jpg'),
(303, 'dbms_book.jpg');
```

```
-- ADDRESS DATA
INSERT INTO Address
(address_id, customer_id, street, city, state, pincode)
VALUES
(1, 101, 'MG Road', 'Lucknow', 'Uttar Pradesh', '226001'),
(2, 102, 'Civil Lines', 'Prayagraj', 'Uttar Pradesh', '211001'),
(3, 103, 'Station Road', 'Kanpur', 'Uttar Pradesh', '208001');

-- ORDER DATA
INSERT INTO Orders
(order_id, customer_id, order_date, total_amount)
VALUES
(501, 101, '2026-08-20', 1498.00),
(502, 102, '2026-08-21', 499.00),
(503, 103, '2026-08-22', 650.00);

-- PAYMENT DATA
INSERT INTO Payment
(payment_id, order_id, payment_date, amount, payment_mode)
VALUES
(601, 501, '2026-08-20', 1498.00, 'UPI'),
(602, 502, '2026-08-21', 499.00, 'Card'),
(603, 503, '2026-08-22', 650.00, 'Cash');

-- DELIVERY DATA
INSERT INTO Delivery
(delivery_id, order_id, delivery_date, status)
VALUES
(701, 501, '2026-08-23', 'Delivered'),
(702, 502, '2028-08-24', 'Shipped'),
(703, 503, '2026-08-25', 'Processing');

-- ORDER ITEM DATA
INSERT INTO Order_Item
(order_id, item_no, product_id, quantity, unit_price)
VALUES
(501, 1, 401, 1, 599.00),
(501, 2, 402, 1, 899.00),
(502, 1, 403, 1, 499.00),
(503, 1, 404, 1, 650.00);
```

Display the Sample data.

```
SELECT * FROM Customer;
SELECT * FROM Seller;
SELECT * FROM Category;
SELECT * FROM Product;
SELECT * FROM Address;
SELECT * FROM Orders;
SELECT * FROM Payment;
SELECT * FROM Delivery;
SELECT * FROM Order-Item;
```

Referential Integrity Violation 1.

Invalid Customer in Address.

The following statement tries to insert an address for a customer who does not exist.

```
INSERT INTO Address
(address_id, customer_id, street, city, state, pincode)
VALUES
(4, 999, 'Main Road', 'Delhi', 'Delhi', '110001');
```

Output

```
ERROR 1452 (23000):
Cannot add or update a child row:
a foreign key constraint fails
```

Reason: customer_id = 999 does not exist in the customer table.

Referential Integrity Violation 2

Invalid Product in Order_Item.

```
INSERT INTO Order-Item
(order_id , item-no , product_id , quantity , unit-price)
VALUES
(501, 3, 999, 1, 500.00);
```

Output :-

```
ERROR 1452 (2300):
Cannot add or update a child row:
a foreign key constraint fails.
```

Reason : product_id = 999 does not exist in the Product table

UNIQUE Constraint Demonstration.

```
INSERT INTO Customer
(customer_id , name , email, phone)
VALUES
(104, 'Rahul', 'aradhana@gmail.com', 9876543213');
```

Output.

```
ERROR 1062 (23000):
Duplicate entry 'aradhana@gmail.com'
for key 'customer.email'
```

Reason : Th email already exists and is declared UNIQUE.

ON DELETE CASCADE Demonstration.

Here, deleting a customer will automatically delete the addresses and orders belonging to that customer.

First check:

```
SELECT * FROM Orders
WHERE customer_id =103;
```

Then execute :

```
DELETE FROM Customer
WHERE customer_id = 103;
```

Now check:

```
SELECT * FROM Orders
WHERE customer_id = 103;
```

Output

Empty Set.

Reason: ON DELETE CASCADE automatically deleted the related child records.

ON DELETE SET NULL Demonstration

Product 401 currently belongs to seller 201.

```
SELECT product_id, product_name, seller_id
FROM Product
WHERE product_id = 401;
```

Output

401	Wireless Mouse	201

Now delete the seller:

```
DELETE FROM Seller
WHERE seller_id = 201;
```

Check the product again:

```
SELECT product_id, product_name, seller_id
FROM Product
WHERE product_id = 401;
```

Output

401 Wireless Mouse NULL

ON DELETE SET NULL keeps the product record but changes its seller_id to NULL.

Final Output.

After inserting valid sample data:

CUSTOMER

101	Aradhana	aradhana@gmail.com	9876543210
102	Riya	riya@gmail.com	9876543211
103	Shatakshi	shatakshi@gmail.com	9876543212

PRODUCT

401	Wireless Mouse	599.00
402	Keyboard	899.00
403	T-Shirt	499.00
404	DBMS Book	650.00

ORDERS

501	101	2026-08-20	1498.00
502	102	2026-08-21	499.00
503	103	2026-08-22	650.00

Conclusion:-

The ER diagram was successfully converted into a relational schema and implemented in MySQL. Primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE, and ON DELETE SET NULL constraints were applied. Sample data was inserted, and referential integrity violations were demonstrated successfully.

27/08/26